

SOTA

Interactive Beat Tracking Annotation GUI

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FEUP 2026

MIR and BT

- **What is MIR?**

- Music Information Retrieval (MIR): research field that focuses on developing computational methods to analyze, interpret and retrieve information from music and audio.
- Beat Tracking (BT): is a field within MIR that aims to computationally detect and follow the temporal positions of beats in music.
- ~~Beat Mapping: process of creating accurate mappings of musical beat along a specific recording or audio file.~~

Beat vs Tactus

- **Beat**
 - A temporal pulse that forms the foundation of a music structure in time, the basis for rhythm.
- **Tactus**
 - It is the human natural feel for beat. It can be characterized by the level at which the listener “taps their foot” to music.

The BT Problem

- Beat Analysis is the basis for many MIR algorithms.
 - *“Many MIR systems first analyze the beats as the starting point, assume the results would be correct, and use them as a unit for further processing.” K. Yamamoto [3]*
- Beat definition can be context based and differ from person to person.
- Beat can vary along a musical piece.

MIR Software

- **Sonic Visualizer (SV)**
 - Open Source (GNU)
 - Designed for precise audio analysis
 - No editing Features

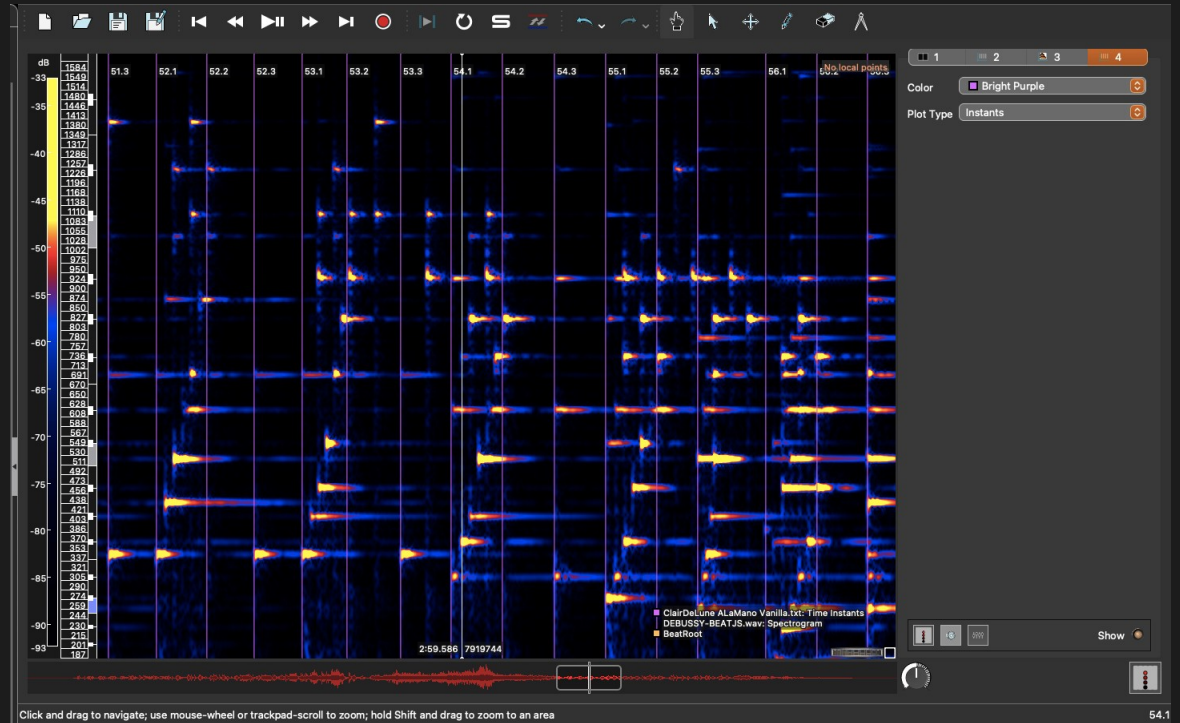


Fig. 1 – Screenshot of Sonic Visualizer v5.2.1

MIR Software

- **Piano Precision [6]**
 - SV Fork
 - Program designed for audio to score alignment.

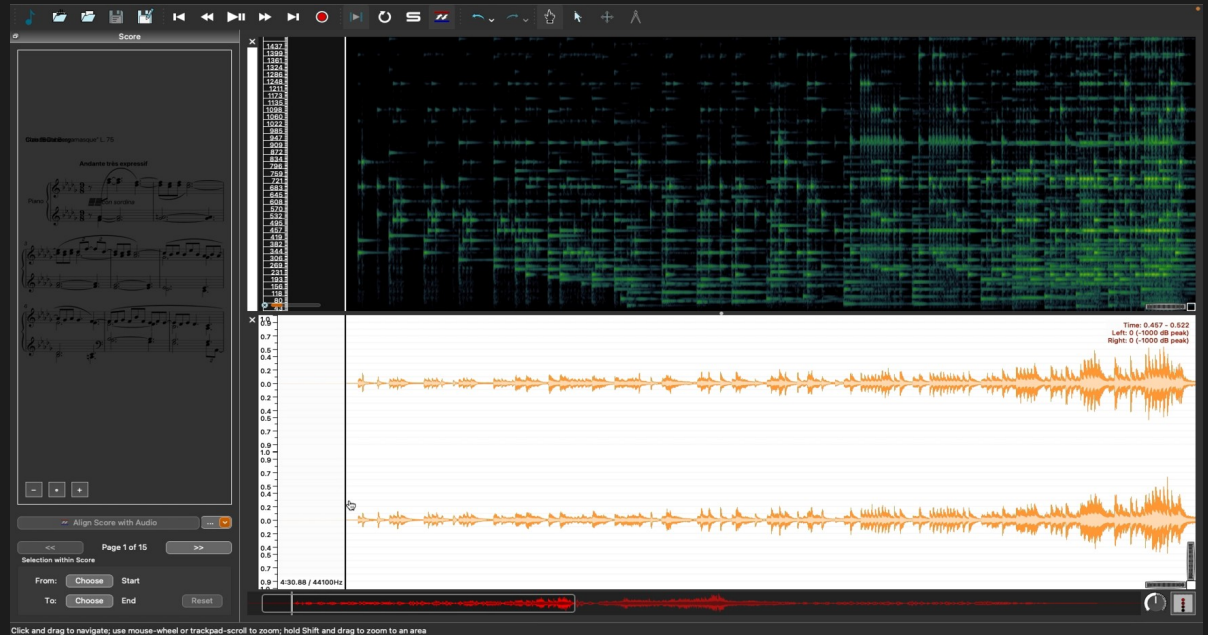


Fig. 2 – Screen-recording of Piano Precision a Sonic Visualizer Fork, v1.1.0
speed x4

BT SOTA

- **Zero-Note Samba (2023) [4]**
 - First Self-supervised BT algorithm.
 - Analyzes **separately** the rhythmic and melodic components of a recording.
 - Works best for music with distinctive percussive instruments (drums) and melodic instruments (vocals, bass, guitar etc...)
- **Beat This! (2024)[5]**
 - Analyzes spectrogram windows of audio to predict beats and downbeats.
 - Works well for a diverse music types and genres.
- **HingeNet (2025)[8]**
 - Works by fusing different ML models and adds a harmonic-aware layer.

- **Songle (2011) [2]**
 - “Songle utilizes music-understanding technologies that automatically estimate music scene descriptions and enable visualization of elements such as structural segments, hierarchical beat structure, melody line, and chords. All users accessing the Songle service can correct estimation errors.” *retrieved from www.songle.jp*
 - Example of a web-interface that enables community collaboration for MIR.
 - Roughly 2 800 000 registered songs.

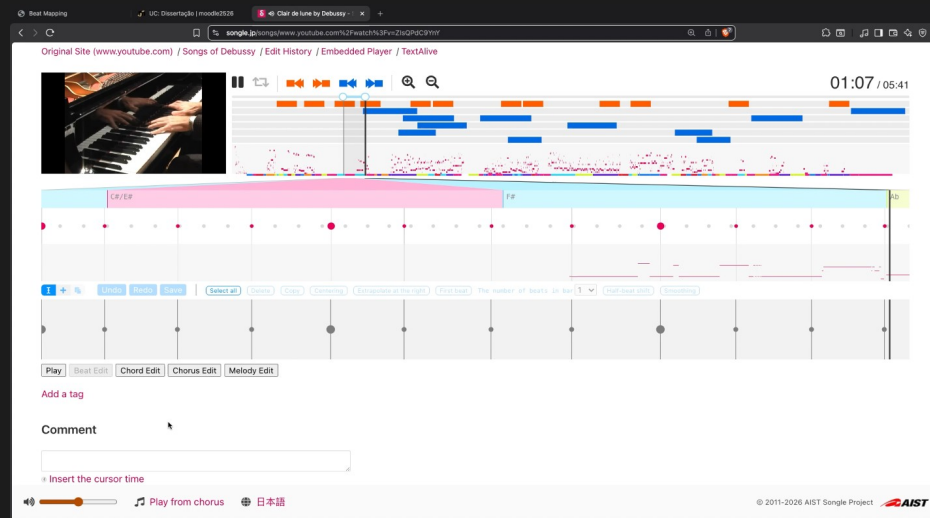


Fig. 3 – Screen-recording of Songle web-interface.

BT Demonstration

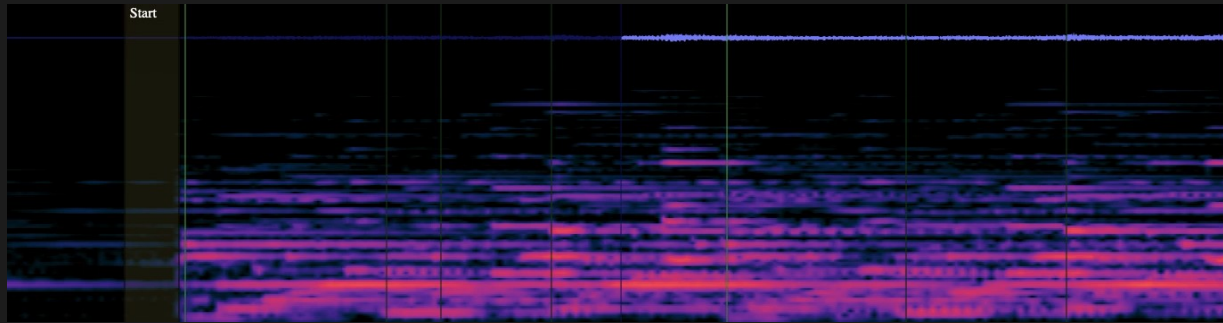


Fig.4 – Screen-recording of an expert from the piece Clair de Lune with BeatThis! unchanged BT Mapping.



Fig. 5 – Excerpt from Clair de Lune.

BT Demonstration

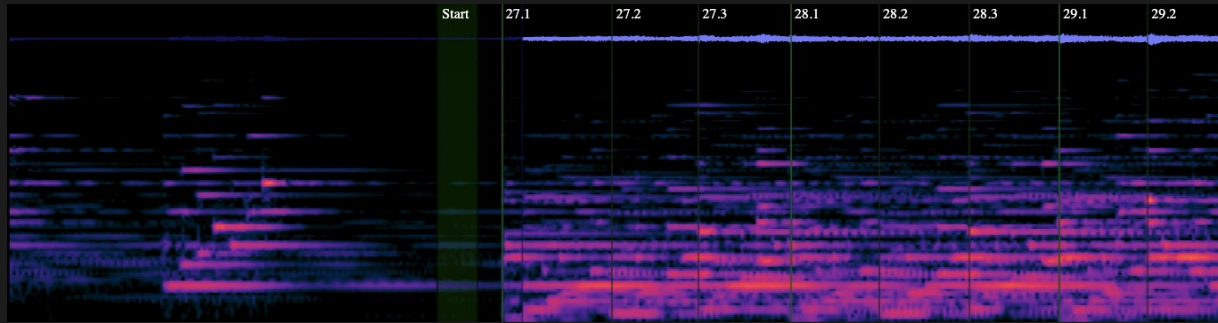


Fig. 6 – Screen-recording of a expert from the piece Clair de Lune with a manually corrected BeatThis! Mapping.



Fig. 7 – Excerpt from Clair de Lune.

Human In the Loop Approach

- Kazuhiko Yamamoto [3]
- HIL Approach for an Interactive BT GUI

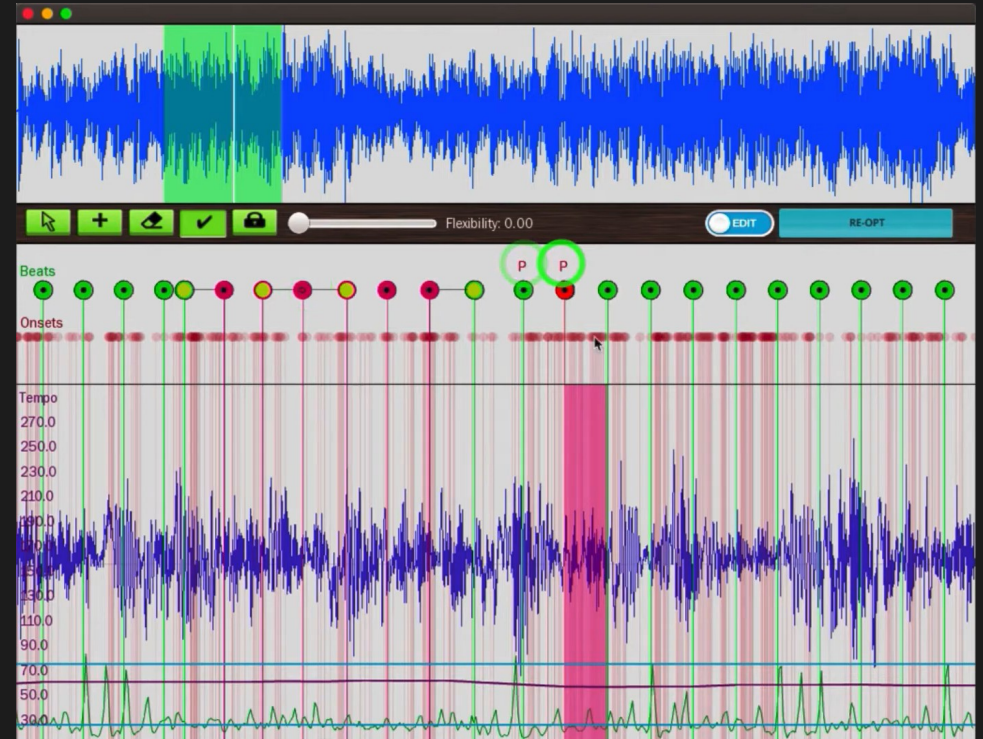
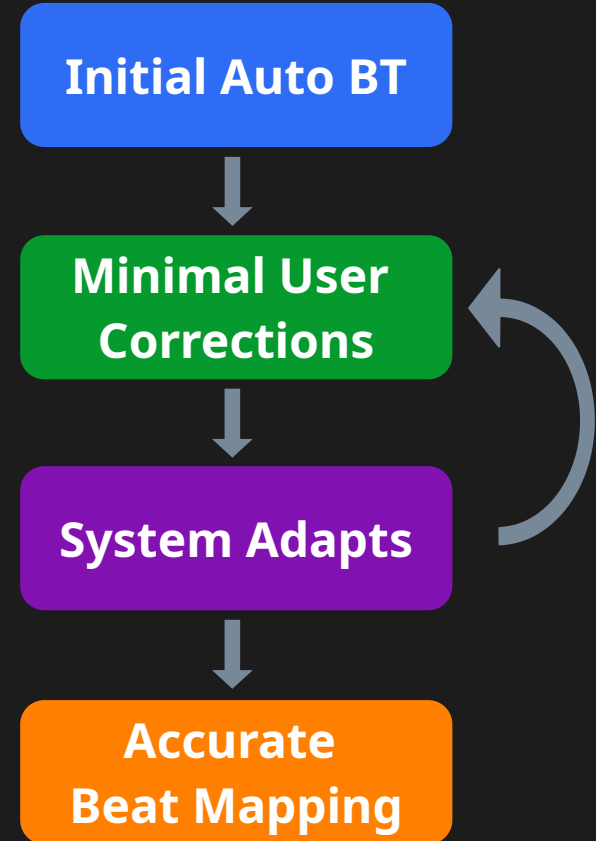


Fig. 7 – Screenshot of K. Yamamoto's HIL BT UI.

Human In the Loop Approach

- Kazuhiko Yamamoto [3]
- HIL Approach for an Interactive BT GUI.
- User makes small corrections, the system adapts.
- Resulting in easy, fast and accurate Beat Mappings.



What does Yamoto's GUI lack?

- No Spectrogram analysis view
- No collaborative features
- No open source repository

Proposed System Direction

- Beat Mapping focused interface
- Integrate SOTA Beat Tracking algorithm/s
- Spectrogram Audio Visualization
- Audio to Score Alignment
- HIL interactive Features
- Collaborative Features
- Open Source codebase

SV Fork or Custom GUI?

Sonic Visualizer Fork

Cons	Pros
Large C++/Qt codebase	Polished visualization and playback Tools
Harder for quick prototyping	Established workflow for SV users
Dealing with legacy code and design decisions.	Large and active user base

Custom GUI

Cons	Pros
Less polished UI/UX	Freedom to decide system architecture
Non-existent initial user base	Faster and easier prototyping
Build audio & visual. Infrastructure from scratch	Ability to introduce modern technologies

References

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- 6) Jiang, Y., Weigl, D. M., & Goebl, W. (2025). "Piano Precision: Advancing Music Performance Analysis by Integrating User-Correctable Audio-to-Score Alignment"
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- 8) Ru, G., Wang, J., Zhao, J., Wu, Y., Yu, Y., Jiang, N., Wang, W., & Li, W. (2025). "HingeNet: A Harmonic-Aware Fine-Tuning Approach for Beat Tracking"



Questions?